c. For runways of 75 feet (22.9 m) in width, the lateral spacing between the inner sides of the rectangular bars centered on the runway centerline is 36 feet (11.0 m).

d. For runways of 60 feet (18.3 m) in width, the lateral spacing between the inner sides of the rectangular bars centered on the runway centerline is 28.8 feet (8.8 m).

Painting Note No. 6

Application of Proportioning Runway Markings for Non-Standard Runway Widths

For runways with widths below 150 feet (45.7 m), the width of the rectangular bars and their lateral spacing between the inner sides of the rectangular bars are adjusted in proportion to the available runway width by using the 150-foot (45.7 m) width runway parameters to determine the percentage decrease in lateral spacing and width of the marking. For example, a nonstandard 70-foot wide runway would apply 70/150 = 0.467 to obtain a lateral spacing for the centered pair of 72 × 0.467 = 33.6 feet (10.25 m), a lateral spacing for other pairs of 5 × 0.467 = 2.3 feet (0.7 m), and a width of 6 feet × 0.467 = 2.8 feet (0.85 m). This adjustment must be such that the inner sides of the innermost rectangular bars to the runway centerline align themselves with the inner side of the aiming point marking. Given that the runway edge markings are painted, the adjustment should be such that the clearance between the runway side strip and the outer edge of the three-bar grouping is a minimum of 1-foot (0.3-m). In no case should the three-bar group be painted farther out from the runway centerline than the aiming point marking.

2.8 Runway Edge Marking.

2.8.1 Purpose.

The runway edge marking provides enhanced visual contrast between the runway edge and the surrounding terrain or runway shoulders, and delineates the width of suitable paved area for runway operations. For lighting provisions, see AC 150/5340-30.

2.8.2 Requirement.

See Table 2-1.

2.8.3 Location.

The runway edge marking consists of two parallel stripes, one placed along each edge of the usable runway with the outer edge of each stripe approximately on the edge of the paved useable runway. For extra wide runways, such as military runways converted for public use, the maximum distance between the outer edges of the parallel stripes is 200 feet (61 m). Figure A-1 illustrates the runway edge marking.

2.8.4 Color.

The stripes of the runway edge marking are white. See paragraph 1.4 for recommended techniques to enhance this marking.

2.8.5 Characteristics.

The runway edge marking has a minimum width of 36 inches (90 cm) for runways of 100 feet (30.5 m) or wider in width and at least 18 inches (45 cm) on smaller width runways (see Table 2-2. for standard runway widths). The runway edge marking extends the full length of the runway as shown in Figure A-7, except for precision runways as shown in Figure A-1 which lack a threshold bar marking.

1. For runways with a displaced threshold, the edge markings continue through the paved area prior to the displaced threshold as shown Figure A-7. This continuation of the edge markings is required because this paved area is used for takeoffs and for landing rollouts from the other direction.
2. Where an aligned taxiway, as shown in Figure A-8 and Figure A-10, precedes a runway threshold, both edge markings will terminate. The point of termination of the edge markings is determined by the taxiway geometry. **Prohibited:** The conversion of a runway section into an aligned taxiway or the construction of a new aligned taxiway is prohibited. See AC 150/5300-13. The FAA further recommends that existing aligned taxiways, for example, as shown in Figure A-8 and Figure A-10, be removed or be reconfigured into usable runways as shown in Figure A-7 or Figure A-11. When a taxiway connects to a runway or the runway has turn pads or turnarounds, the runway edge marking remains continuous between the runway and these adjoining infrastructures.
3. For intersecting runways, see guidance in paragraph 2.2.

2.9 Runway Threshold Bar Marking and Runway Displaced Threshold Marking.

The complete marking scheme for a runway with a displaced threshold, when required by AC 150/5300-13, includes a runway threshold bar and arrowheads with and without arrow shafts. Figure A-7, Figure A-10, and Figure A-11 illustrate the various applications of displaced thresholds and the requirement for additional marking components. For lighting provisions, see AC 150/5340-30.

2.9.1 Runway Threshold Bar Marking.

2.9.1.1 **Purpose.**

The runway threshold bar marking serves two functions by delineating the start of the (i) displaced threshold by indicating the beginning section of the runway available for landing from the unusable section on the approach side of the displaced threshold as shown in Figure A-7, Figure A-10, and Figure A-11 and (ii) regular threshold from prepared surfaces before the runway that are used for a blast pad, stopway, EMAS or end of an aligned taxiway as shown in Figure A-8 and Figure A-9.

2.9.1.2 **Requirement.**

Install a runway threshold bar marking when there is a displaced threshold, blast pad, stopway, EMAS, or aligned taxiway on the approach side of the threshold.

2.9.1.3 **Location.**

The runway threshold bar marking is an elongated rectangular bar that is located perpendicular to the runway centerline and on the landing portion of the runway. The outboard edge of the marking is aligned with the location labeled "start of runway displaced threshold" as shown in Figure A-10 or "start of runway threshold" as shown in Figure A-9. Note that the entire runway threshold bar marking is on the portion of the runway available for landing.

2.9.1.4 **Color.**

The runway threshold bar marking is white. See paragraph 1.4 for required techniques to enhance this marking.

2.9.1.5 **Characteristics.**

The runway threshold bar marking is 10 feet (3.1 m) in width and extends between the runway edges or between the runway edge markings.

2.9.2 Arrow Marking.

The arrow marking (arrowheads with and without arrow shafts; depending on the length of the displacement) performs three possible functions, that is, two cases for displaced thresholds and one case for a runway threshold with an aligned taxiway.

2.9.2.1 **Purposes.**

1. Figure A-7 illustrates the predominant case in which the threshold is displaced from the runway end. In this case, white arrowheads with and without arrow shafts are required to identify the portion of the runway before the displaced threshold to provide centerline guidance for pilots during approaches, takeoffs, and landing rollouts from the opposite direction.
2. Figure A-10 illustrates the rare case in which a displaced threshold is preceded by an aligned taxiway. In this case white arrowheads with and without arrow shafts and yellow arrowheads without arrow shafts are required to identify the runway portion from the aligned taxiway portion. Furthermore, a yellow runway demarcation bar is required to identify the start of the runway. See paragraph 2.9.3 for runway demarcation bar criteria and figures.
3. Figure A-8 illustrates the rare case in which a runway threshold (not displaced) is preceded by an aligned taxiway. In this case, yellow arrowheads without arrow shafts are required to identify the runway portion from the aligned taxiway portion.

Safety Notice No. 4

14 Code of Federal Regulation (CFR) Part 139 Certificated Airports Runway Incursion Mitigation Requirement

Extended runway holding position markings, as illustrated in Figure D-4, provide a safety benefit to pilots of ADG V and VI aircraft that lowers the risk of a runway incursion. Uniformity of surface markings contributes to airfield safety by providing visual instructions that result in consistent interpretation by pilots. Application of this surface painted marking, which is part of the taxiway centerline marking standard under 14 CFR Section 139.311(a)(2), establishes compliance with 14 CFR Part 139.

3.3.3 Color.

The Pattern A marking scheme, as shown in Figure A-13, is yellow and, when painted on light-colored pavements, outlined in black. See paragraph 1.4 and Table 1-1 for required techniques to enhance this surface marking.

3.3.4 Characteristics.

3.3.4.1 Pattern A for the runway holding position marking consists of a set of two continuous lines, two dashed lines, and three spaces that are all parallel, extend across the entire width of the taxiway, measure 12 inches (30 cm) in width, and are separated as shown in Figure A-13. It is located laterally such that a dash is centered on the taxiway centerline. Where the marking extends unbroken over more than one taxiway centerline, locate the marking such that a dash is centered on one taxiway centerline. At airports that do not have an operating ATCT and are not certificated under Part 139, the width and separation measurement may be reduced from 12 inches (30 cm) to 6 inches (15 cm). For clarification, all airports certificated under Part 139, whether or not there is an operating ATCT, are required to use the 12-inch measurement.

3.3.4.2 The taxiway centerline marking is interrupted by the runway holding position marking as shown in Figure A-13.

3.3.4.3 For taxiways having taxiway edge markings, interrupt the taxiway edge marking as shown in Figure A-23 so that the Pattern A marking continues to the edge of the defined taxiway width. For taxiways that connect to runways that serve Airplane Design Groups (ADGs) V or VI airplanes as defined by AC 150/5300-13, the marking is further extended onto both paved shoulders (similar to Pattern A, as shown in Figure D-4) except when the Pattern A surface marking governs over the Pattern B surface marking per paragraph 3.4.2.1. For both airplane design groups, the length of the marking from the taxiway centerline onto the paved shoulder measures 62.5 feet (19 m). The 62.5-foot (19-m) measurement takes into account the downward viewing angle from the cockpit while the pilots are